

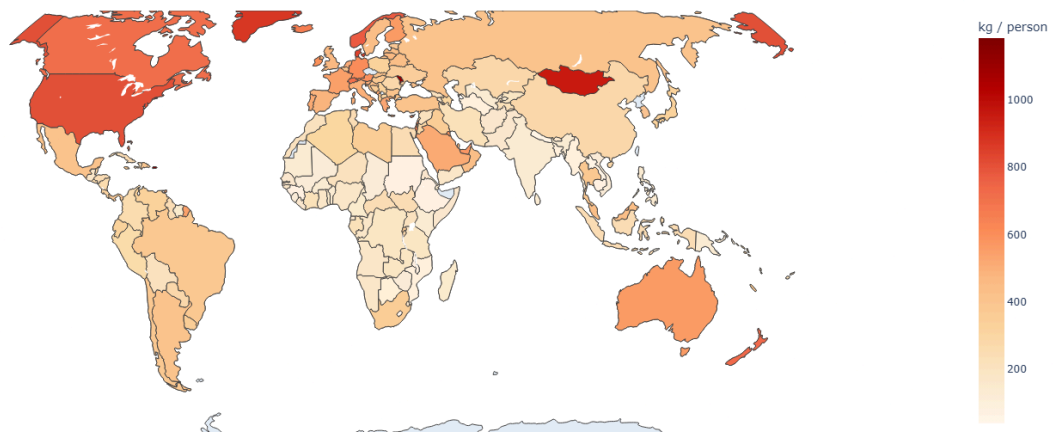
Data Analysis Report

Subject:	Sem-V - E Waste & Environmental Mgt.
Course Teacher:	Prof. Nilima Main
Group No:	14
Name of the students:	Vishal Prajapati - 24101B0047 Altamash Ansari -24101B0048 Meet Alshi -24101B0049
Problem Statement:	Population vs E-Waste: Is More People = More Waste? Examine correlation between population and e-waste generation.
Dataset Source:	https://www.kaggle.com/datasets/veetikoivuniemi/united-nations-sdg-goal-11-waste-dataset

DashBoard Link:	https://ewaste-dashboard-vishal47.streamlit.app/
Collab Notebook link:	https://drive.google.com/file/d/13L1-vZBMSzBCru2y-jkqulFAnZFuY3mx/view?usp=sharing

Visualization 1:

Global E-Waste Generation Per Capita (kg/person)



WHAT THE GRAPH SHOWS

The map reveals sharp spatial inequality: the highest `wastepercap` values cluster in wealthy Western Europe, North America, and small high-income island territories, while the lowest values sit almost entirely across Sub-Saharan Africa and South/Southeast Asia. Country **size on the map has no visible relationship to color intensity** — some of the largest-population countries (India, Nigeria, Indonesia) sit in the lightest color band.

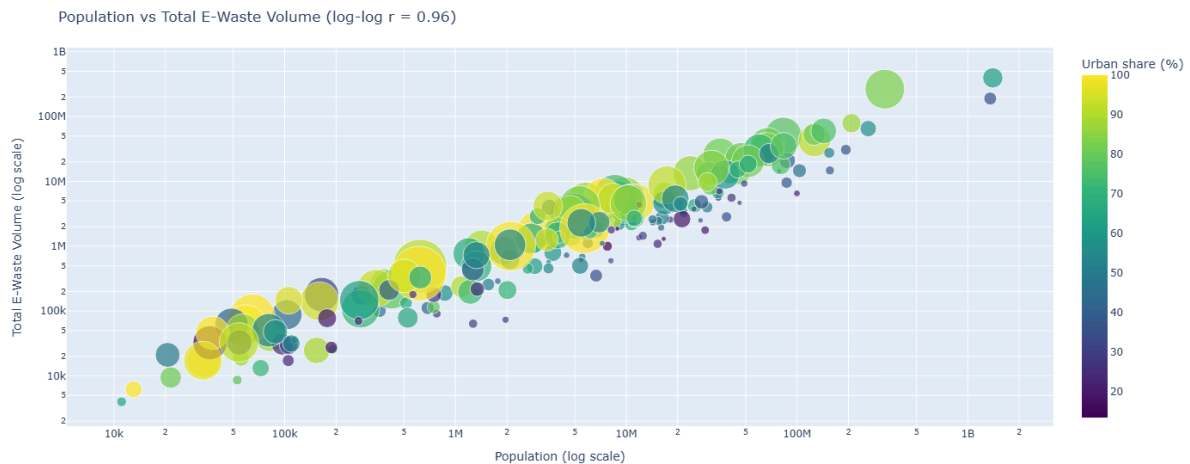
ROOT PROBLEM IDENTIFIED

Per-person e-waste intensity is being driven by purchasing power and consumption habits, not by population pressure. High-income populations replace electronics on short upgrade cycles, concentrating disposal risk regardless of how many people live there.

TARGETED RECOMMENDATION

Enact right-to-repair legislation, restrict planned obsolescence practices, and mandate minimum extended-warranty periods in the highest-intensity (dark red) regions identified on the map — the policy lever here is consumption behavior, not population control.

Visualization 2:



WHAT THE GRAPH SHOWS

Population and total national e-waste volume are strongly positively correlated (linear $r \approx 0.88$, log-log $r \approx 0.96$). Points fall close to a straight line on the log-log axes — a near-proportional relationship. Bubble size (GDP) adds nuance: some large-population, lower-GDP bubbles (India, Indonesia, Nigeria) still post enormous *total* volumes purely from scale, even before wealth effects are considered.

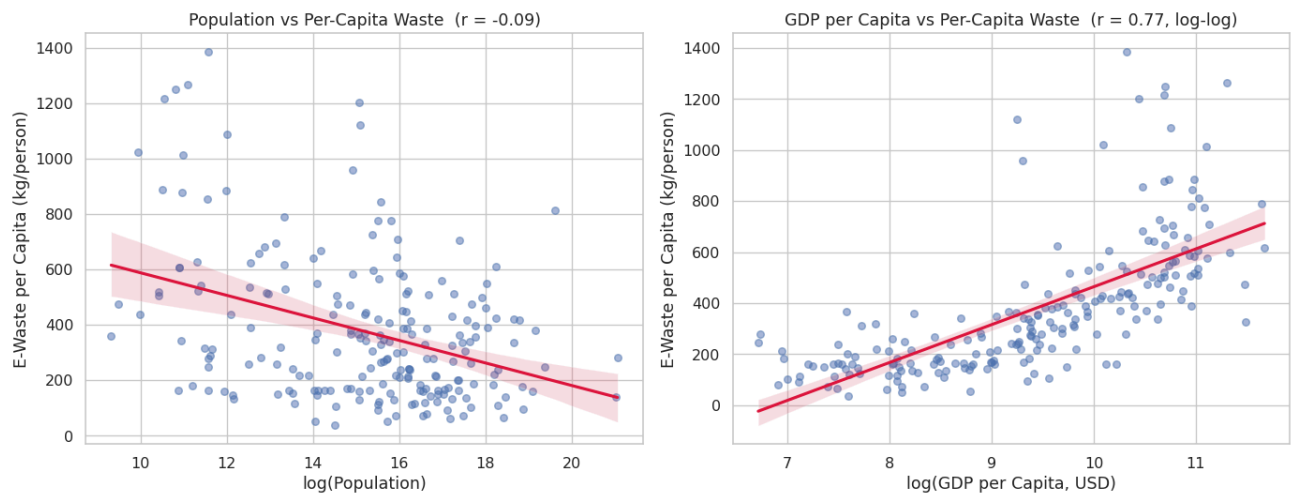
ROOT PROBLEM IDENTIFIED

In large, rapidly growing developing nations, sheer volume — not per-person excess — overwhelms local collection and recycling infrastructure. Without formal channels, this waste is frequently handled through open-air burning and informal, hazardous recycling, exposing communities to lead, mercury, and brominated flame retardants.

TARGETED RECOMMENDATION

Build formal municipal e-waste collection grids and industrial-scale recycling hubs specifically in high-population, high-total-volume, developing countries — sized to absolute throughput, not per-capita affluence.

Visualization 3:



WHAT THE GRAPH SHOWS

The left panel is nearly flat — population has essentially **no correlation** with per-capita e-waste ($r \approx -0.09$). The right panel shows a clear upward trend — GDP per capita is a strong predictor of per-capita waste (linear $r \approx 0.67$, log-log $r \approx 0.77$). Side by side, the same dataset tells opposite stories depending on which denominator you use.

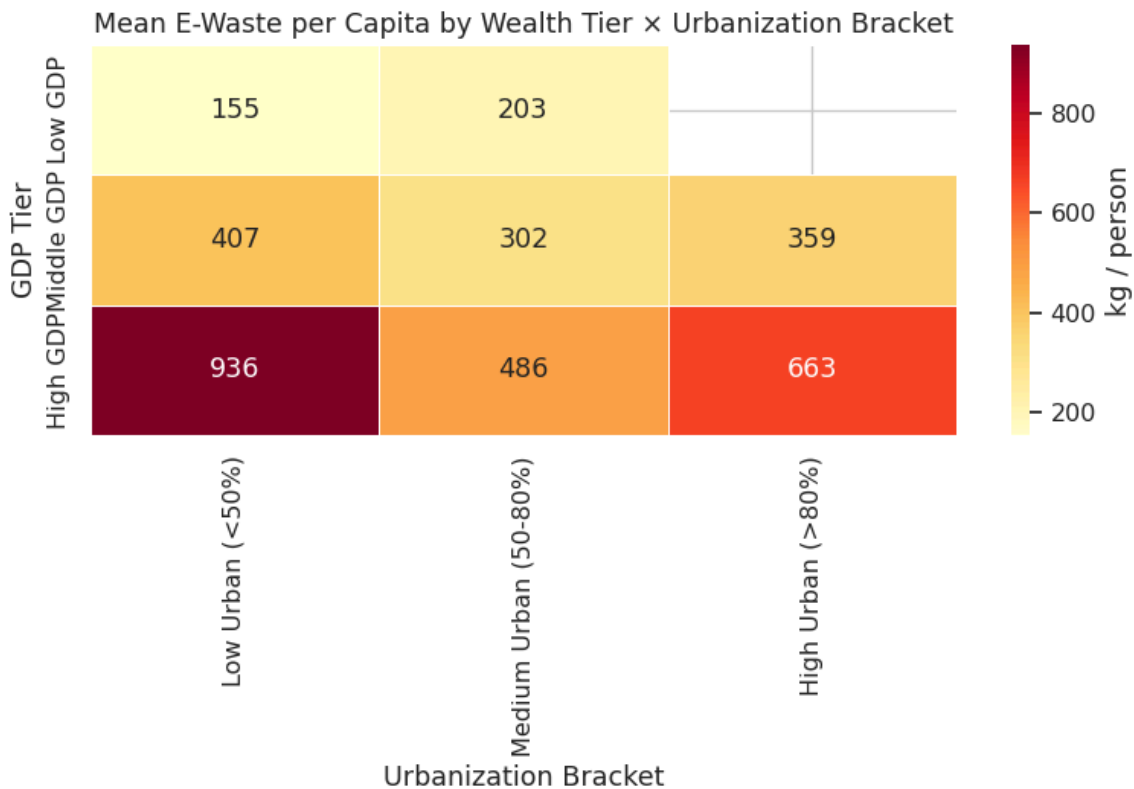
ROOT PROBLEM IDENTIFIED

High per-capita purchasing power accelerates device turnover — short 18–24 month upgrade cycles for smartphones and consumer electronics — and wealthy markets also absorb transboundary e-waste flows exported from their own disposal streams into poorer regions.

TARGETED RECOMMENDATION

Enforce Extended Producer Responsibility (EPR) mandates in high-GDP markets, requiring manufacturers to fund take-back, buy-back, and certified recycling logistics proportional to the devices they sell — targeting the wealth-driven behavior directly, not population size.

Visualization 4:



WHAT THE GRAPH SHOWS

Wealth tier is the dominant driver: mean per-capita waste rises from ~170 kg/person (Low GDP) to ~342 kg/person (Middle GDP) to ~620 kg/person (High GDP), regardless of urbanization bracket. Urbanization shows a real but secondary effect and interacts unevenly with wealth — some high-GDP, lower-urbanization cells (small sample, oil-and-resource economies) post the single highest values, worth flagging as a small-n outlier rather than a general urban-density pattern.

ROOT PROBLEM IDENTIFIED

High-income, urbanized populations concentrate large volumes of discarded electronics into dense geographic zones, straining municipal drop-off capacity even where wealth alone would predict manageable per-person volumes.

TARGETED RECOMMENDATION

Mandate retail take-back and trade-in programs, and deploy automated smart e-waste collection kiosks in dense urban centers within high-GDP tiers — targeting the convenience bottleneck where wealth and density compound.

Conclusion & Final Verdict

Executive Summary Matrix

Visualization	Key Finding	Root Problem	Recommendation
Choropleth Map	Per-capita waste is spatially concentrated in wealthy regions, independent of population size	Consumption intensity tracks income, not headcount	Right-to-repair laws; ban planned obsolescence
Log-Log Bubble Scatter	Population strongly predicts total national volume ($r \approx 0.88$ linear, 0.96 log-log)	Mass volume overwhelms infrastructure in large developing nations	Build formal collection & recycling hubs at national scale
Faceted Regression	Population has ~no link to per-capita waste ($r \approx -0.09$); GDP does ($r \approx 0.67-0.77$)	Wealth-driven short device life cycles	Extended Producer Responsibility (EPR) mandates
Wealth × Urban Heatmap	Per-capita waste rises sharply with GDP tier; urban density compounds it	Urban convenience bottleneck in high-income cities	Retail trade-in programs; smart collection kiosks

Direct Answer

"Does more people mean more e-waste?"

YES for national volume - NO for individual impact.

- More people **does** mean more total national volume: population sets the *baseline scale* a country must manage ($r \approx 0.88$ linear, 0.96 log-log with total waste).
- More people **does NOT** mean higher individual waste intensity ($r \approx -0.09$ with per-capita waste). A person in a high-GDP country generates roughly **5–6x more e-waste** than a person in a low-GDP country, independent of that country's population.
- **Final synthesis:** population dictates the *scale of infrastructure* a country needs to build; economic wealth drives the *personal consumption behavior* that determines how much waste each individual actually creates.